National University of Computer and Emerging Sciences
Islamabad Operating System Lab Spring 2025

Lab Task 14

Task 1: Limited Simultaneous Food Orders in a Restaurant

Scenario:
A restaurant has a limited number of cooking stations (e.g., 4 stations). Multiple customers place food orders, but the kitchen can only handle a certain number of orders simultaneously. The remaining orders must wait for a station to become available.

Objective:
Use counting semaphores to control the number of orders being cooked at the same time, ensuring that no more than 4 orders are processed at once. The remaining orders must wait for an available station.

Task 2: Library Study Room Allocation

Scenario:
A library has ten study rooms. Students arrive and request a room to study.

Objective:
Use a counting semaphore to manage the allocation of

study rooms, allowing up to ten students to study simultaneously.

Task 3: Ticket Booking System

Scenario:
A cinema has a limited number of seats (e.g., 50), and multiple customers are trying to book tickets for the same show. Each customer needs a seat, and the system must ensure that no more than 50 tickets can be booked at the same time.

Objective:
Use counting semaphores to manage the available tickets, ensuring that customers can only book a ticket if there is availability, and they must wait if the maximum limit is reached.

Task 4: Traffic Light Simulation at an Intersection

Scenario:
At a busy intersection, there are two sets of traffic lights for two perpendicular roads. Each light can either be green or red, but only one set can be green at any time to avoid collisions.

Objective:
Use a counting semaphore to ensure that the lights for the two roads are never green simultaneously, providing safe and orderly traffic flow.

Task 5: Shared Memory Access for Multiple Applications

Scenario:
Multiple applications need access to a shared memory space for reading and writing data. However, only one application can write to the memory at a time, and multiple applications can read simultaneously.

Objective:
Use counting semaphores to manage access to the shared memory, ensuring that no more than one writer is allowed at a time while allowing multiple readers simultaneously.

THE END
